

List of derivative shortcut rules

Here is the list of shortcut rules. Check them off as you use them. Keep this list for reference!

☐ sum rule: $\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$

☐ constant multiple rule: $\frac{d}{dx}[c \cdot f(x)] = c \cdot \frac{d}{dx}[f(x)]$

☐ constant function rule: $\frac{d}{dx}[c] = 0$

☐ power rule: $\frac{d}{dx}[x^n] = nx^{n-1}$

☐ exponential function rule: $\frac{d}{dx}[a^x] = a^x \cdot \ln(a)$

trigonometric function rules:

☐ $\frac{d}{dx}[\sin(x)] = \cos(x)$

☐ $\frac{d}{dx}[\cos(x)] = -\sin(x)$

☐ $\frac{d}{dx}[\tan(x)] = \sec^2(x)$

☐ product rule: $\frac{d}{dx}[f(x) \cdot g(x)] = \frac{d}{dx}[f(x)] \cdot g(x) + \frac{d}{dx}[g(x)] \cdot f(x)$

☐ quotient rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x) \cdot \frac{d}{dx}[f(x)] - f(x) \cdot \frac{d}{dx}[g(x)]}{[g(x)]^2}$

☐ chain rule: $[f(g(x))]' = f'(g(x)) \cdot g'(x)$

☐ logarithmic function rule: $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$

Just for reference, here are a few more that come up sometimes. They get used much less and they don't show up on this problem set; I would just be sad if I didn't write them down.

• inverse trigonometric function rules:

◦ $\frac{d}{dx}[\arctan(x)] = \frac{1}{1+x^2}$

◦ $\frac{d}{dx}[\arcsin(x)] = \frac{1}{\sqrt{1-x^2}}$

• rules for other trigonometric functions:

◦ $\frac{d}{dx}[\cot(x)] = -\csc^2(x)$

◦ $\frac{d}{dx}[\sec(x)] = \sec(x) \tan(x)$

◦ $\frac{d}{dx}[\csc(x)] = -\csc(x) \cot(x)$